

Karthik Raja Rasamsetti

QA Automation Engineer · AI-Driven Quality Engineering

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SUMMARY

QA Automation Engineer / SDET with 3+ years shipping scalable UI, API, database, and performance test automation, and architecting Unified Test Automation Frameworks (UTAF) that cut regression cycles and lifted coverage. Early adopter of AI-driven Quality Engineering, building agentic systems with LangGraph, LangChain, and Graph RAG to generate, orchestrate, and triage tests.

TECHNICAL SKILLS

Languages	Java, JavaScript, Python, SQL
Automation	Selenium, Playwright, WebdriverIO, Appium
Frameworks	TestNG, Cucumber (BDD), Mocha, Pytest, RestAssured
Performance	k6, WebLOAD
CI/CD	Azure DevOps, GitHub Actions, GitLab CI, CircleCI
Tools	Jenkins, Docker, Postman, Git
AI in QE	Agentic RAG, Graph RAG, LangChain, LangGraph, Prompt Engineering
Database & Methods	MySQL, PostgreSQL, Agile / SDLC / STLC

EXPERIENCE

Performance & QA Automation Engineer — Feuji

2022 — Present

QA & AI Quality Engineering delivered across client engagements — logistics, fintech/insurance, and event management.

OnTrac · Logistics & Transportation

2024 — Present

- Designed and executed the performance testing strategy with k6, holding system stability through peak holiday traffic (~2.5x normal load).
- Migrated legacy WebLOAD performance tests to the open-source k6 CLI, eliminating commercial licensing costs and modernizing the performance testing framework.
- Integrated the k6 test suite with Azure DevOps pipelines for automated CI/CD execution and Grafana dashboards for real-time performance monitoring, visualization, and trend analysis.
- Built an end-to-end Pytest framework covering UI flows and database validation across 4 service modules.
- Integrated automation pipelines into Azure DevOps for PR-gated continuous testing — reduced manual regression effort ~60%.

Tech: k6, WebLOAD, Python, Pytest, Azure DevOps

LeaseLock · FinTech / Insurance

2023 — 2024

- Architected UTAF (Unified Test Automation Framework) standardising UI, API, and DB testing across teams — adopted by 3 squads.
- Built reusable, modular components covering end-to-end policy and underwriting workflows.
- Wired suites into CircleCI with automated failure reporting and nightly summaries; lifted coverage ~50% with fewer flaky failures.

Tech: Selenium, Playwright, JavaScript, Java, CircleCI, UTAF

Encore · Event Management

2022 — 2023

- Developed and maintained BDD automation in Cucumber for event-management workflows (hotels, infrastructure, services).
- Authored cross-browser suites running on Chrome, Firefox, Safari, and Edge in parallel.
- Collaborated in Agile ceremonies and partnered with developers on shift-left QA practices.

Tech: Selenium, Cucumber (BDD), Java, TestNG, Cross-browser (Chrome/Firefox/Safari/Edge)

AI QUALITY ENGINEERING

QualiBrix — AI-powered test automation platform (Feuji product)

- Built a Graph RAG service enriching Jira story context with verified ground truth from a knowledge graph, eliminating LLM guesswork and improving AI-generated test-case quality.
- Contributed to platform design and architecture, driving key technical decisions with the team; worked on the Test Management Tool service.
- Troubleshoot and resolved multi-tenant architecture issues across the RAG services; tested all platform services, surfacing and resolving severe defects through rigorous QA.

Tech: Python, LangGraph, Graph RAG, KùzuDB, FastAPI, PostgreSQL, Multi-tenant Architecture

Personal / Open-Source Projects

- MediClear — full-stack health app that reads prescriptions and lab reports and explains them in plain language across English, Telugu, and Hindi, with read-aloud, trend charts, and a safety-first "always consult your doctor" design. Multi-provider LLM vision layer (Gemini/Claude/OpenAI) behind a factory pattern; FastAPI + React, JWT auth, per-user history.
- qa-engine — an 11-agent LangGraph platform turning a plain-English user story into a validated, Docker-sandboxed, self-healing Playwright test with a stakeholder report.
- diagnostician — a read-only agent that triages CI/CD failures into application-bug / broken-test / flaky / environment, with confidence scoring and recommended action.
- Prototyped Agentic RAG for test-data generation and LangChain pipelines for test-case generation from acceptance criteria; applied prompt engineering to speed script authoring ~30%.

Tech: Python, LangGraph, LangChain, Playwright, Docker, FastAPI, React, Gemini, OpenAI, Anthropic, ChromaDB, KùzuDB

KEY ACHIEVEMENTS

- Authored UTAF — now standardising test automation across multiple LeaseLock products.
- Made performance testing release-blocking before peak business windows, protecting stability at ~2.5× load.
- Cut regression cycle time and lifted automation coverage through framework optimisation and CI/CD parallelisation.
- Pioneered AI-driven Quality Engineering at Feuji — from Graph RAG test-case grounding to multi-agent test generation and triage.

EDUCATION

B.Tech, Electronics & Communication Engineering

Jun 2018 — Jul 2022

Ideal Institute of Technology, Vidyut Nagar, Kakinada · Grade: A